

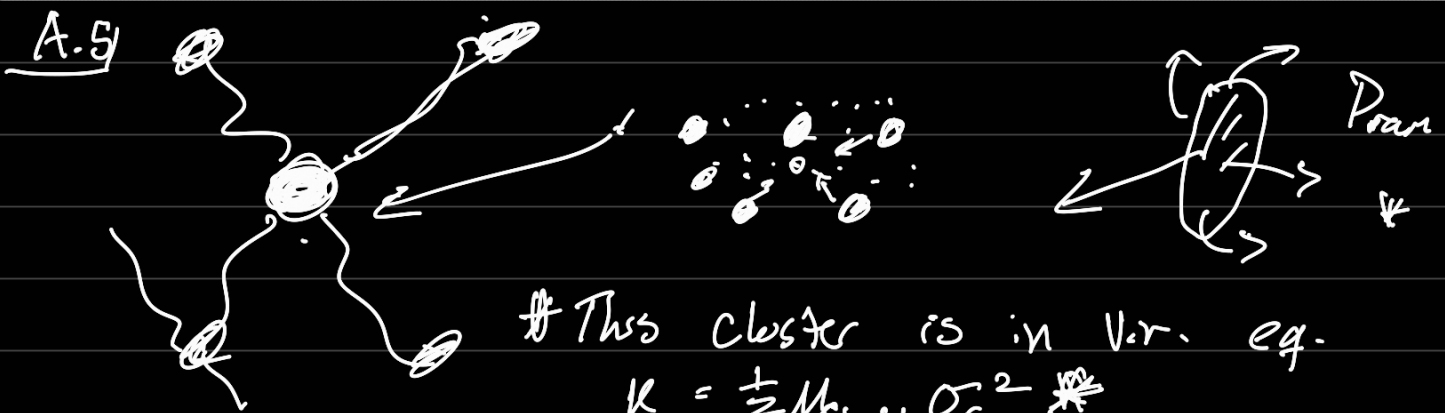
B.3] $\dot{R}^2 = \frac{8\pi G}{3} \rho_v R^2 - kc^2$

• $k=0$: $R(t) \propto e^{Ht}$
 • $k=1$: $R(t) \propto \frac{\cosh(Ht)}{\sinh(Ht)} = \frac{e^{Ht} + e^{-Ht}}{e^{Ht} - e^{-Ht}}$ *
 • $k=-1$: $R(t) \propto \frac{\cosh(Ht)}{\sinh(Ht)} = \frac{e^{Ht} + e^{-Ht}}{e^{Ht} - e^{-Ht}}$ *

* $1/2 = \frac{8\pi G}{3} \rho_v R^2 - kc^2$ *
 $t \rightarrow \infty$

(b) $\Omega = \rho_v / \rho_c$
 $\rho_c = \frac{3H^2}{8\pi G}$
 $1/2 = \frac{8\pi G}{3} \rho_v - kc^2 = \frac{8\pi G}{3} \Omega \rho_c - kc^2$
 $H^2 = H^2 \Omega - kc^2$
 $\frac{H^2 + kc^2}{H^2} = \Omega(t)$
 $H^2 = \frac{\dot{R}^2}{R^2}$
 $1 + \frac{kc^2 R^2}{\dot{R}^2} = \Omega(t)$

(c) $H = \frac{\dot{R}}{R} = \frac{t^{-1/2}}{2t^{1/2}} \sim \frac{1}{2t}$ *
 (d) $\sim e^{t/2t_c}$
 Radi: $\sim (t/t_0)^{1/2}$



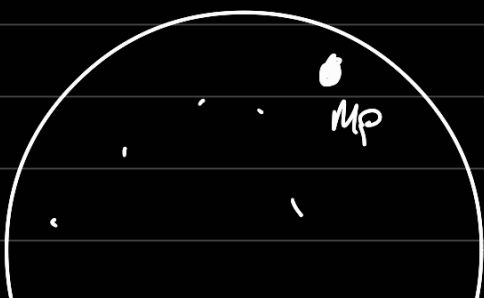
* This cluster is in Vir. eq.

$K = \frac{1}{2} M_{clust} \sigma_c^2$ *

$U = - \frac{GM_c m_p}{R}$

$K = \frac{3}{2} K T$

$3KT = GM_c m_p$



* $2K + U = 0$

$\Rightarrow$ Argued about $p^+ \rightarrow e^-$, loses half K.E.

$$T_{true} = \frac{1}{2} t_0 = \frac{GM_{comp}}{6k_B R}$$

$$T_{true} = \frac{GM_{comp}}{6k_B R}$$

* Primerdual: F_c ✓

$$E = \epsilon m c^2, \quad U = -\frac{GM^2}{R}, \quad R_{sch} \\ \epsilon = \frac{1}{2} \\ = \frac{1}{2} \frac{1}{2} \frac{1}{3} \quad \hat{U} = \frac{1}{2} U \quad *$$

less from shortcut $R_{sch} \rightarrow$ last stable orbit
 $3R$

$\rightarrow .1$

A.6 | $\rho_c: K=0$

$$\rho_c = \frac{3H^2}{8\pi G}$$

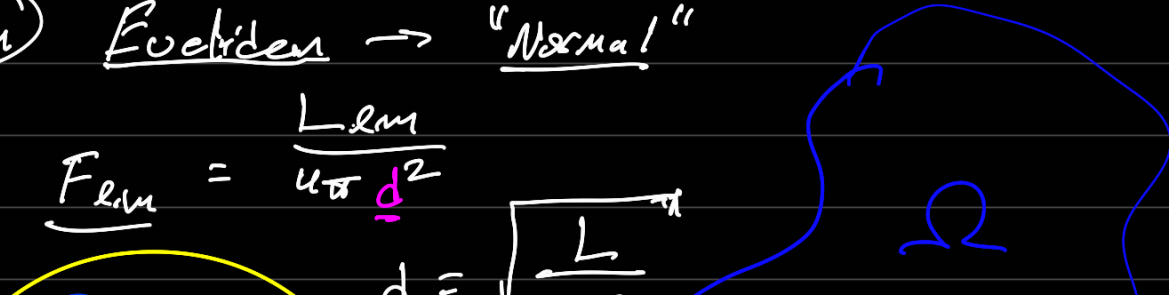
$$\Omega: \quad \Omega_i = \frac{\rho_i}{\rho_c} = \frac{\rho_i 8\pi G}{3H^2}$$

A.7 | Come back $\leftarrow$ + any thing "dunno"

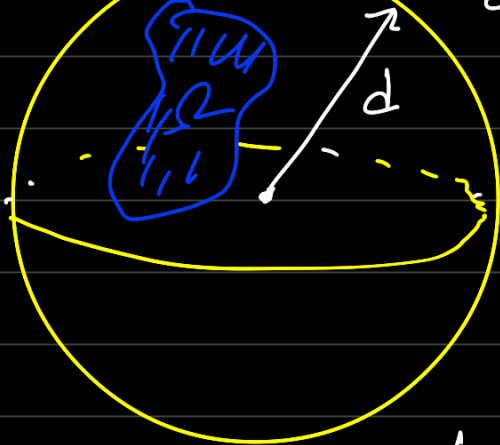
A.8 | yesterday

B.2 | (a) Euclidean $\rightarrow$ "Normal"

above $F_{em} = \frac{L_{em}}{4\pi d^2}$

* 

$$d = \sqrt{\frac{L}{\Omega}}$$



$$Vol = \frac{4\pi}{3} d^3$$

$$Vol = \frac{4\pi}{3} \left(\frac{h}{4\pi F_{em}} \right)^{3/2} = \frac{4\pi}{(4\pi)^{3/2} 3} \left(\frac{L}{F_{em}} \right)^{3/2}$$

✱

Sphere: S.A. = $4\pi R^2$

→ Redians: unit-sphere: $R=1$ S.A. = 4π

$$\frac{\Omega}{4\pi} \leftarrow \text{total}$$

$$(Vol)_{true} = \left(\frac{\Omega}{4\pi} \right) (Vol) = \boxed{\frac{\Omega}{3(4\pi)^{3/2}} \left(\frac{L}{F_{em}} \right)^{3/2}}$$

$\downarrow$ $\rightarrow$ $\rightarrow$
 V_{true} $\rightarrow$ $\rightarrow$

ϕ : luminosity function
similar to IMF:

$$\phi(L) dL : \frac{\# \text{ Gal}}{Vol} \cdot W/L \text{ between } L \text{ and } L+dL$$

$$n = \int \xi(M) dM \neq$$

$$n(L) = \int \phi(L) dL$$

$$N = \int (Vol)_{true} \phi(L) dL$$

IMF: $\xi(M)$ Role: "Count" stars in Gal mass of those stars
"Count" α :

$$\textcircled{1} N = \int \xi(M) dM$$

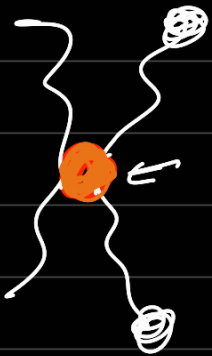
Lum function: ϕ : Role: ① Count the # of galaxies
w/ luminosities L + dL

$$N = \int Vol \phi(L) dL$$

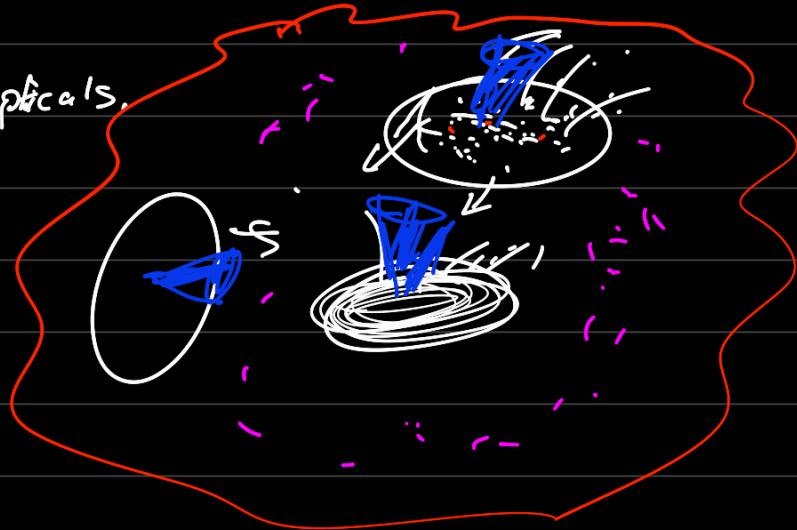
$$n = \int \phi(L) dL$$

A.41

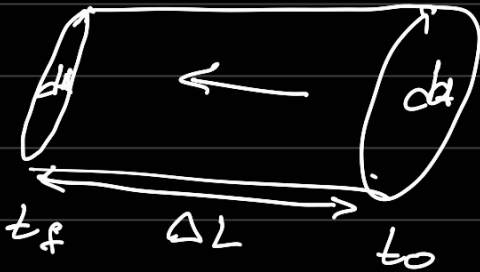
- In Gal. clusters: mostly Ellipticals.
- Towards edges you find spirals
- Spirals on "web"



$$F = \frac{dP}{dt}$$



$$p = mV$$



$$Vol = dA \cdot L$$

$$\rho_{ISM} = \frac{M}{Vol} = \frac{M}{dA \cdot L} \Rightarrow \boxed{M = \rho_{ISM} dA \Delta L}$$

$$\Delta F = \frac{\Delta P}{\Delta t} = \frac{M \Delta V}{\Delta t} = \frac{\rho_{ISM} dA \Delta L \Delta V}{\Delta t}$$

$$\frac{\Delta F}{dA} = P = \frac{F}{A} = \rho_{ISM} \left(\frac{\Delta L}{\Delta t} \right) \Delta V$$

$$\frac{\Delta L}{\Delta t} = \Delta V$$

$$P = \rho_{ISM} \Delta V^2$$

$$P \approx \rho_{\text{ISM}} v^2$$

$$P = \beta \rho_{\text{ISM}} v^2$$

A.6.1

(a) $\rho_m = \frac{M}{\text{Vol}}$
matter

$\rho_w = R^{-2(1+w)}$ *

as the universe expands $\frac{dM}{dt} = 0$ con. of mass/Eu.

$$\rho_m = \frac{M}{R^3} \Rightarrow$$

$$\rho_m \propto R^{-3}$$

$$E = mc^2 \quad c=1 \quad E=m$$

(b) $\rho_r = n_\gamma E_\gamma = n_\gamma \propto \frac{1}{R^3}$

$$E_\gamma = h\nu = \frac{hc}{\lambda}$$

$$\lambda = R \left(\frac{\lambda_0}{R_0} \right) \nearrow$$

* $\frac{\lambda_0}{R_0} = \frac{\lambda}{R}$ *

$$E_\gamma \sim \frac{1}{R}$$

$$\rho_r = n_\gamma E_\gamma \sim \frac{1}{R^3} \cdot \frac{1}{R} \sim \frac{1}{R^4}$$

$$\rho_r \propto R^{-4}$$

(c) $\rho_v = \text{const.} \leftarrow$ Vacuum states don't change

A.7.1 looking for kT relationship

(a) Ultra-rel: $kT \gg mc^2$ $E = pc$ *

$$n = \frac{g}{(2\pi\hbar)^3} \int_0^\infty \frac{4\pi p^2 dp}{e^{pc/kT} \pm 1}$$

$$y = \frac{pc}{kT} \Rightarrow p^2 = \left(\frac{kT}{c} \right)^2 y^2$$

$$1 \quad kT \quad 1$$

$$\boxed{dp = \frac{u}{c} dy}$$

$$n = \frac{u \sigma g}{(2\pi\hbar)^3} \int_0^\infty \frac{\left(\frac{\hbar}{c}\right)^2 y^2 \left(\frac{\hbar}{c}\right) dy}{e^y \pm 1}$$

$$= (kT)^3 \left[\dots \right]$$

⑥ non-rel: $kT \ll mc^2$, $E = \frac{p^2}{2m} + mc^2$

$$n = \frac{u \sigma g}{(2\pi\hbar)^3} \int_0^\infty \frac{p^2 dp}{e^{p^2/2mkT} e^{mc^2/kT} \pm 1}$$

$$kT \ll mc^2 \Rightarrow mc^2/kT \gg 1$$

$$e^{mc^2/kT} \gg 1$$

$\Rightarrow$ Ignore ± 1

$$n \approx \frac{u \sigma g}{(2\pi\hbar)^3} e^{-mc^2/kT} \int_0^\infty e^{-p^2/2mkT} p^2 dp$$

$$y^2 = p^2/2mkT \quad \text{or} \quad y = p/\sqrt{2mkT}$$

$$\boxed{dy \sqrt{2mkT} = dp}$$

$$\approx e^{-mc^2/kT} \left(\frac{u \sigma g}{(2\pi\hbar)^3} \right) \int_0^\infty e^{-y^2} (2mkT) y^2 (\sqrt{2mkT} dy)$$

$$n \approx \underbrace{(2mkT)^{3/2}} e^{-mc^2/kT} \left(\dots \right)$$

⑦ Depending on model:

$$e^{-mc^2/kT} \sim 1 - \frac{mc^2}{kT} + \frac{1}{2} \left(\frac{mc^2}{kT} \right)^2 - \dots$$

• make a cut off:

i) $\frac{mc^2}{kT} \leftarrow \nrightarrow$

ii) $\left(\frac{mc^2}{kT} \right)^5 \leftarrow$

$$A.8) T dS = dE + PdV \quad \& \quad dS = 0 \leftarrow$$

• Closed universe so micro states $\uparrow$ Entropy
aren't changing for the grand canonical ens.

$$0 = dE + PdV$$

Assume:

(i) $E = mc^2$

(ii) $V = \frac{4\pi}{3} R^3$

$\rho = \frac{m}{V}$

$$dE + PdV = 0$$

divide by dt

WRONG...

$$\frac{dE}{dt} + P \frac{dV}{dt} = 0$$

$$\frac{d}{dt} \left(\frac{\rho}{V} c^2 \right) + P \frac{dV}{dt} = 0$$

See extra notes

$$\left(\frac{\dot{\rho}}{V} \right) c^2 - \frac{\rho}{V^2} \dot{V} c^2 + P \dot{V} = 0$$

$$\dot{V} = \frac{4\pi}{3} \frac{d}{dt} R^3 = 4\pi R^2 \dot{R}$$

$$V \left(\frac{\dot{\rho}}{V} c^2 - \frac{\rho}{V^2} c^2 (4\pi R^2 \dot{R}) + P 4\pi R^2 \dot{R} \right) = 0 \quad \checkmark$$

$$\dot{\rho} c^2 - \frac{\rho}{V} c^2 (4\pi R^2 \dot{R}) + P 4\pi R^2 \dot{R} = 0$$

$$\dot{\rho} c^2 - \frac{\rho}{\frac{4\pi}{3} R^3} c^2 (4\pi R^2 \dot{R}) + P 4\pi R^2 \dot{R} = 0$$

$$\dot{\rho} c^2 - \frac{3\rho}{R} \dot{R} + P 4\pi R^2 \dot{R} = 0$$

$$\dot{\rho} c^2 - \frac{3\dot{R}}{R} \rho - P (4\pi R^2 \dot{R}) \left(\frac{4\pi}{3} R^3 \right)^{-1} = 0$$

$$\uparrow \left(\frac{1}{V} \right) \quad \uparrow V \rightarrow \frac{1}{V}$$

$$\dot{\rho} c^2 - \frac{3\dot{R}}{R} \rho = P \frac{4\pi R^2 \dot{R}}{\frac{4\pi}{3} R^3} = 0$$

$$P \frac{3\dot{R}}{R}$$

$$\dot{\rho} c^2 - \frac{3\dot{R}}{R} \rho c^2 - P \frac{3\dot{R}}{R} = 0$$

$$c^2 \dot{\rho} + \frac{3\dot{R}}{R} (\rho c^2 + P) = 0$$